PHYS 143 – Mechanics and Thermal Physics

Orientation

How does this course work, and what must you do this week to be ready for the physics?

Hope Donglo

Group E • Office 24, Department of Physics, University of Ghana

Part 1 of 2: Orientation and the course outline (Tuesday 27 October)

First semester 2026/2027 • BSc Level 100 • 3 credits

Welcome

Who is teaching, what is the course,
and what can you do by the end?

KEY IDEA

PHYS 143 is a 3-credit course in mechanics and thermal physics, taught to Group E by Hope Donglo.

GROUP E FACILITATOR
Hope Donglo
Room 24, Department of Physics
hdonglo@ug.edu.gh
Office hours: Wednesdays, 10:00 to 12:00
Personal page: hhdonglo.github.io/phys143.html

3 credits	**8** groups
None prerequisite	**None** teaching assistants

Quick poll: Who is already logged in to Sakai?

All information and course contents are found in Sakai: a must visit site

Open PHYS 143 on Sakai: sakai.ug.edu.gh

What this course is about

Understand how the physical world works and learn the mathematical tools needed to predict it.

Understanding nature
Mechanics: motion, forces, energy and momentum
Thermal physics: temperature, heat and thermodynamic processes

Mathematical tools
Vectors and calculus for physical modelling

Prediction
Predict and solve real physical problems

Physical world	Mathematical tools
Why cars stop	Vectors
Why projectiles curve	Calculus
	Problem solving
Why bridges expand	
	Quantitative reasoning
How engines produce work	

EACH WEEK Read → Attend → Practice

By the end of this contact you can

1. **Name** the reference text and where slides, notes and videos are posted
2. **Log in** to Sakai and **select a group** with no clash
3. **State** how you are assessed, and the homework and quiz rules
4. **State** when and where Group E meets, and the office hours

What you will study

Now, what does the semester cover?

PHYS 143 is a 3-credit, calculus-based introduction to mechanics and thermal physics for Level 100 science students.

- **Mechanics:** vectors; motion in one, two and three dimensions; Newton's laws; work, energy and power; momentum, impulse and collisions; rotation and gravitation.
- **Thermal physics:** temperature and heat; thermal properties of matter and the kinetic theory of gases; the first and second laws of thermodynamics.
- **Purpose:** to train you to describe physical phenomena, model them mathematically, and predict what happens from physical laws. For the first time in your science career you will solve problems with vectors and calculus, and build the confidence to tackle problems that at first seem difficult.
- **Relevance:** these ideas underpin later physics courses and are used across the physical and medical sciences and engineering.

Discover the outcomes

Before I show you the outcomes, try to work them out from one everyday event.

A car brakes to a stop on a wet road. How far does it travel?

- **What do you need to describe it?** Quantities with units and direction: speed, time, distance, force.
- **How would you model it?** With equations: vectors for direction, calculus for changing speed.
- **How would you predict the stopping distance?** With physical laws: Newton's second law and the work-energy relation.

The abilities this course builds

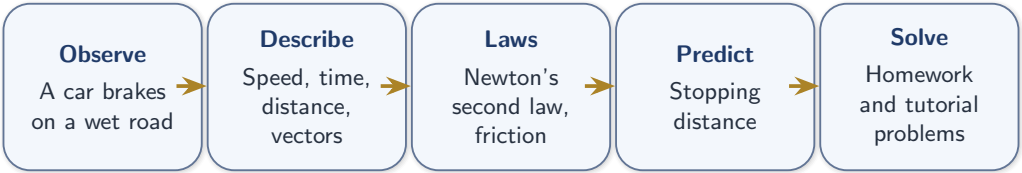
Describe physical phenomena, model them mathematically, and predict from physical laws.

Here are the same abilities in the course's own words, with where each is tested.

No.	You will be able to	Tested in
1	**describe** motion with vectors and units, and **model** it with calculus	homework, interim, final
2	**apply** Newton's laws to **predict** the motion of a system	homework, interim, final
3	**analyze** systems with work, energy and momentum conservation, including collisions	homework, interim, final
4	**model** rotation with angular quantities and **explain** Kepler's three laws	Homework 6, final
5	**explain** and **predict** thermal processes with the ideal gas, kinetic theory and the first and second laws	Homework 7, final
6	**evaluate** problems independently from textbooks and online sources, and **apply** physics to other sciences	homework: reading, work on time

How every topic is built

Every chapter follows the same path. Here it is with the braking car.



So where does the learning happen? In class and on Sakai.

Face to face

- Two lecture periods a week in your group, with worked examples and short quizzes.
- Tutorials to discuss the course material and homework, at times and venues set by your group facilitator.

Online on Sakai

- Lecture videos, slides and notes, uploaded every week.
- Homework problem sets and the interim assessment.
- Announcements, the discussion forum and your marks.

Other online methods, such as live sessions on Microsoft Teams or Zoom, are used when a face-to-face meeting cannot hold.

Your weekly routine and ground rules

A steady weekly routine makes this work. Here it is, with the ground rules.

Each week you will

1. Do the assigned reading in Young and Freedman before class.
2. Attend your group's lectures and tutorial, where short quizzes may be given.
3. Work through the homework on Sakai and submit it by its due date.

Ground rules

- Attend lectures and tutorials in your own group.
- Submit homework on the day it is due; late homework is not recorded.
- Quizzes cannot be made up, so come to class prepared.
- Check Sakai every week for materials, announcements and deadlines.

Course map and chapters

What do we study, and in which week?

KEY IDEA

Every week has a chapter to read before the lecture, and homework follows.

Course map and chapters

Week	Lecture	Topic	Chapter and sections to read	Notes
Week 0 Introduction	none	Orientation; physical quantities	Chapter 1: sections 1.1 and 1.3 to 1.6	Add/drop closes 6 Nov
Week 1	1	Vectors	Chapter 1: sections 1.7 to 1.10	none
Week 2	2	Motion in one, two and three dimensions	Chapters 2 and 3: whole	Homework 1 sent
Week 3	2	Motion, continued	Chapters 2 and 3: whole	Homework 2 sent
Week 4	3	Newton's laws and applications	Chapter 4: whole. Chapter 5: sections 5.1 to 5.4	Homework 3 sent
Week 5	4	Work and kinetic energy	Chapter 6: whole	Homework 4 sent
Week 6	5	Potential energy and conservation	Chapter 7: whole	Homework 5 sent

Interim assessment: Saturday 5 December 2026, on Sakai, Weeks 1 to 5 topics. The University interim period opens 23 November.

Course map and chapters

Week	Lecture	Topic	Chapter and sections to read	Notes
Week 7	7	Momentum, impulse, collisions	Chapter 8: sections 8.1 to 8.5	Reading assignment starts
Week 8	9	Temperature and heat	Chapter 17: whole	Reading continues
Week 9	10	Thermal properties of matter	Chapter 18 except 18.4; also 17.5	Homework 6 sent
Christmas break: 24 December to 3 January				
Week 10	11	First law of thermodynamics	Chapter 19: whole; also 18.4	Homework 7 sent
Week 11	12	Second law of thermodynamics	Chapter 20: whole	Teaching ends 15 Jan
Week 12	none	Revision and tutorials	none	Course evaluation closes 22 Jan
Week 13	none	Final examination	All content	Date set by Academic Affairs

Self-study: rotation and gravitation

Lecture	Topic	Chapter and sections to read
8	Rotation of rigid bodies; dynamics of rotation; Kepler's laws	Chapter 9: whole. Chapter 10: sections 10.1 to 10.6. Chapter 13: Kepler's laws and gravitation

Not taught in class.
Notes are on Sakai. Tutorials take your questions.

Homework 6 checks this reading, and the material is examined.

Reference texts

Which books and resources will you use?

KEY IDEA

Young and Freedman, University Physics, is the required text.

REQUIRED

Young and Freedman

University Physics, 15th edition

Chapters 1 to 8, 9, 10, 13, 17 to 20

Copies: Physics Library and Balme Library

Additional reading

Halliday, Resnick and Walker

Tipler

Serway and Jewett

Giancoli

Backhouse and Houldsworth: mathematics

Bring to class: a scientific calculator and a notebook for worked examples

Free resources and simulations

PhET simulation	Topic
Projectile Motion	Motion in two dimensions
Energy Skate Park	Work and energy
Collision Lab	Momentum and collisions
Gravity and Orbits	Gravitation (reading)
Gas Properties	Gases and heat

OpenStax University Physics
Free online and as PDF. Volume 1: mechanics. Volume 2: thermodynamics.

Sakai

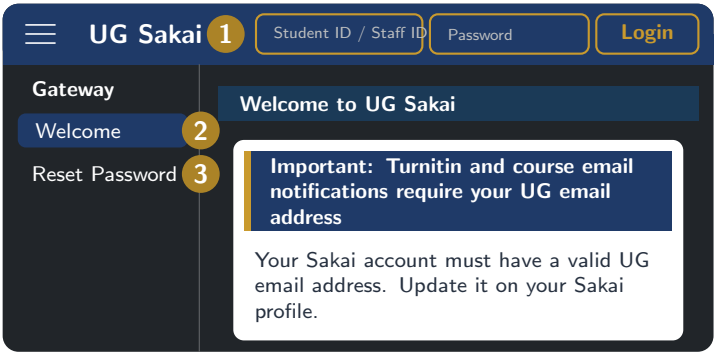
Where does everything happen online?

KEY IDEA

Sakai is the course website: videos, slides, notes, homework and marks.

Sakai: logging in

Step 1



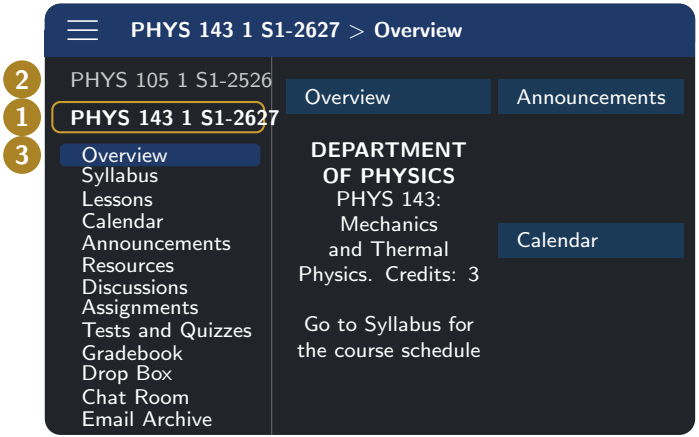
1 Log in with your Student ID and password

2 Read the Welcome notice

3 Forgot your password? Use Reset Password

Add your UG email address to your Sakai profile,
or you may miss course emails and Turnitin may fail

Sakai: finding PHYS 143



1 Open **PHYS 143 1 S1-2627** in your site list

2 PHYS 105 1 S1-2526 is a separate site, online this week. Check it too.

3 Start on Overview. The Syllabus has the schedule.

Sakai tour: the left menu

13 tools

Tool	Use
Overview	Course home page
Syllabus	Outline and schedule
Lessons	Weekly content
Calendar	Class and deadline dates
Announcements	Schedule changes
Resources	Videos, slides, notes
Discussions	Ask questions

Tool	Use
Tests and Quizzes	Homework and interim
Gradebook	Your marks
Drop Box	Private files with your facilitator
Email Archive	Messages to the site email
Assignments	Tasks to submit
Chat Room	Live chat with the class

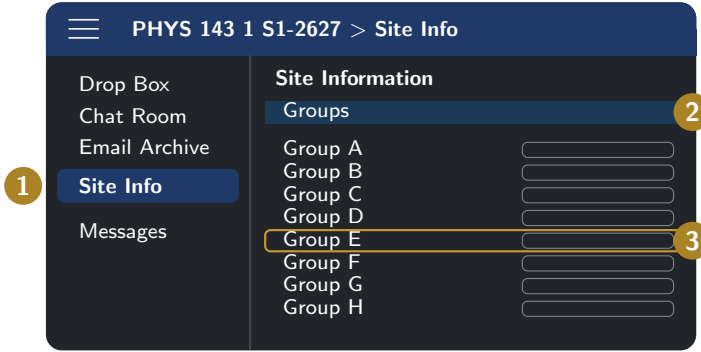
Predict first: which two tools will you open most often? Compare at the end of the semester.

All groups at a glance

Compare all eight timetables before you pick a group, so that you avoid a clash.

Group	Lecture period 1	Lecture period 2
A	Wednesday 14:30 to 15:20, KAB	Thursday 07:30 to 09:20, KAB
B	Monday 11:30 to 13:20, N1	Thursday 07:30 to 09:20, NNB3
C	Monday 15:30 to 17:20, NNB2	Thursday 11:30 to 13:20, NNB2
D	Tuesday 07:30 to 09:20, JQB24	Wednesday 07:30 to 08:20, KAB
E	**Tuesday 09:30 to 11:20, JQB19**	**Wednesday 07:30 to 09:20, N3**
F	Monday 09:30 to 11:20, KAB	Wednesday 08:30 to 09:20, KAB
G	Tuesday 07:30 to 09:20, KAB	Wednesday 15:30 to 16:20, KAB
H	Monday 17:30 to 19:20, L/OT1	Thursday 07:30 to 09:20, N1

KAB: K.A. Busia Building. N1, N3: N-N Block. NNB2, NNB3: New N Block. JQB19, JQB24: Jones Quartey Building. L/OT1: Larway/Oracca Tetteh Building.



1 Open **Site Info** in the left menu

2 Find the groups available to join

3 Join the group that fits your timetable, with no clash. Click **Join**.

Deadline: it will be announced on Sakai

Try not to choose a group that clashes. If every group clashes, there is no solution.

Assessment

How is your work marked?

KEY IDEA

Homework and quizzes 10%, interim assessment 30%, final examination 60%.

How you are assessed

100%

10%

Homework and quizzes

Seven sets on Sakai, and short quizzes in lectures

30%

Interim assessment

Multiple choice, on Sakai

60%

Final examination

Multiple choice, all content



Assessment	Rule
Homework	Submit **on the day it is due**. Marked for correct answers with correct units. Due days and times will be announced.
Quizzes	Short questions on reading and recent lectures. The format, length and split of the 10% will be discussed.
Interim, 30%	Make-up only with a doctor's notice. Follow the Sakai announcements for the start time.
Final, 60%	Set by Academic Affairs: date, time and venue
Polls and exit checks	Not marked, but they count toward class participation, which can earn a grade. Follow the Sakai announcements for its weight.

Late homework is not recorded. No make-up quiz is given.

Academic integrity

Academic integrity means honesty, fairness, responsibility, respect and accountability in academic work.

Situation	Rule
Homework and all work you submit	It must be **your own work**
Another student's answers	Copying is **misconduct**
Quizzes, interim, final examination	**No AI tools** unless a facilitator allows
Studying with an AI tool	**Name the tool** and how you used it

Read the **UG Plagiarism Policy** and the **UG Policy on the Use of Artificial Intelligence**

Contact and support

When and where do we meet, and where do you get help?

KEY IDEA

Group E meets on Tuesday and Wednesday;
office hours are on Wednesday, 10:00 to 12:00.

When and where, and getting help

Session	Day and time	Venue
Tuesday lecture	From 27 Oct (orientation), 09:30 to 11:20	JQB19
Wednesday lecture	From 28 Oct (orientation), 07:30 to 09:20	N3
Office hours	Wednesday 10:00 to 12:00. Secondary: yields to the PHYS 105 lab from 18 Nov	Room 24
Tutorial	To be discussed	

If you	Then
Have a question	Email your facilitator: hdonglo@ug.edu.gh
Have a technical problem	Tell your facilitator at once
Have special needs	Tell Dr Joseph Asare this week.

Missed a lecture or a change of schedule? Check **Sakai** first

You know when and where we meet. Here is the rest of the help available.

- **Catch up:** lecture videos, slides and notes are on Sakai every week.
- **Questions between classes:** the Sakai discussion forum, or email your group facilitator.
- **Feedback:** homework and quiz marks are posted on the Sakai Gradebook.
- **Textbooks:** copies of the required text are in the Physics Library and the Balme Library; free open textbooks are on the Resources slide.
- **Online access:** do homework and the interim on a computer with a reliable connection. If a technical problem stops a submission, tell your facilitator straight away.
- **Missed interim for health reasons:** a make-up is possible with a doctor's notice signed by a University physician.
- **Special needs:** tell the Course Coordinator this week; see the UG policy on Students and Staff with Special Needs.

Your roadmap to success

Support only helps if you use it. Here is the weekly plan.

Step	What you do	What it earns	Outcomes
1	Read the assigned chapter before class	you can answer quiz questions on the reading	all
2	Attend lectures and work the examples	quiz marks, and a clear picture of the method	1 to 3
3	Review slides and notes on Sakai	you fill the gaps from class	1 to 5
4	Submit homework on its due day, with units	the 10% for homework and quizzes	2 to 5
5	Bring your hard problems to the tutorial	worked solutions before the tests	2 to 5
6	Ask early: office hours, the Sakai forum, or email	you fix small problems early	all

Your roadmap to success

That is one week. Now the whole semester.

When	Milestone	Outcomes
Week 0, Introduction	join Sakai, choose a group; matriculation Sat 31 Oct; add/drop closes 6 Nov	none
Weeks 1 to 6	Homework 1 to 5, quizzes; University interim period opens 23 Nov	1 to 3
Saturday 5 Dec	interim assessment on Sakai, covering Weeks 1 to 5 (30%)	1 to 3
From Week 7	reading: rotation and Kepler; Homework 6 (Week 9)	4 and 6
24 Dec to 3 Jan	Christmas break	none
Weeks 10 to 12	Homework 7; teaching ends 15 Jan; evaluation 11 to 22 Jan; revision	5
23 Jan to 14 Feb	final examination, multiple choice, all content (60%)	1 to 5

The schedule may change at short notice; lost lectures are made up at times announced on Sakai.

Close

What do you do next?

KEY IDEA

Log in to Sakai, choose your group, and read Chapter 1, sections 1.1 and 1.3 to 1.6.

1. **Log in** to Sakai and open PHYS 143 1 S1-2627
2. **Choose your group** before the deadline, with no clash
3. **Read** Chapter 1, sections 1.1 and 1.3 to 1.6, before Wednesday's lecture

Exit check

1. Which assessment is worth 60%, and in what format?
2. What happens to homework submitted after the due day?
3. Where will you find this week's slides and notes?

Answer online on Sakai.

That is the outline. If anything is unclear, this is how to reach me.

GROUP E FACILITATOR

Hope Donglo

Room 24, Department of Physics
hdonglo@ug.edu.gh
Office hours: Wednesdays, 10:00 to 12:00

Personal page: hhdonglo.github.io/phys143.html

End of Orientation

Next lecture: Physical Quantities and Uncertainties (Wednesday 28 October)

All course materials are posted on Sakai.

Part 2 of 2 – Wednesday 28 October

Physical Quantities and Uncertainties

How do we describe nature with numbers, and how far can we trust them?

Hope Donglo

Group E • Office 24, Department of Physics, University of Ghana

By the end of this lecture you can

That was the orientation. Now the physics begins, with measurement. Here is what you should be able to do by the end.

1. **Convert** quantities to SI units and check that equations are dimensionally consistent.
2. **State** a measurement with its uncertainty and the right significant figures
3. **Round** the result of a calculation correctly

Here is the route for today, so you can see why each step follows from the one before.

1. Measurement
Physics needs **measurement**.

2. Quantities
What we measure are **physical quantities**.

3. Types
Quantities are **fundamental** (independent) or **derived** (defined from fundamental ones).

4. Units
Every quantity needs a **unit**.

5. Standards
Units need **standards**: the SI.

6. Using quantities
Check equations, convert units, and state the uncertainty of every measurement.

Physics and measurement

Why does physics need measurement, and what is a physical quantity?

KEY IDEA

A physical quantity is a measurable property of a system or phenomenon that can be expressed by a numerical value and a unit.

We start at the first step: why does physics need measurement at all?

Experimental science

Physics is an **experimental science**.

Observe and find patterns

Physicists **observe** the phenomena of nature and look for **patterns** that relate them.

Laws, principles, theories

Patterns are called **theories** or, when well established, **laws** or principles. A theory explains phenomena from observation.

Rule: status of a theory

No theory is ever the ultimate truth. New observations may require it to be revised or discarded. We can disprove a theory, but never prove it always correct.

Galileo

Free-fall acceleration is independent of weight if the force from the air (resistance and buoyancy) is much less than the weight. In a vacuum a feather and a cannonball fall together.

Galileo's claim could be tested only because falling can be measured.

Observation
A car brakes on a wet road.

Quantity
Speed and stopping distance are chosen.

Measurement
Distance and time are read with instruments.

Unit
Values are stated in m, s and m/s.

Model
A law or equation relates the quantities.

Prediction
The stopping distance at a new speed.

Rule: a description that cannot be measured cannot be checked against experiment, so measurement sits in the middle of the path.

Words depend on the speaker
"The car is fast." Fast compared with what?

Numbers can be checked
"The car travels at $25\ \mathrm{m/s}$." Anyone with the instruments can check it.

Physics describes quantities. Mathematics shows how they are related.

Algebra: relationships

An equation links quantities. For the braking car, speed = distance ÷ time.

Vectors: direction

Some quantities need a direction as well as a size, such as the velocity of the car.

Calculus: change

It describes how a quantity changes, such as how the speed of the car changes with time.

Graphs: patterns

A graph of measured values shows the pattern, such as distance against time for the car.

Every one of these tools works on quantities with a value and a unit, so we start with the quantities.

Physics needs things that can be measured. What do we call them, and what exactly are they?

Definition: physical quantity

A **physical quantity** is a measurable property of a system or phenomenon that can be expressed by a numerical value and a unit.

Physical meaning

It is a property we can compare with a standard. Weight and height are two physical quantities that describe you.

Unit

The numerical value means something only with its unit. The unit names the standard used for the comparison.

Example

Length, **time**, **mass**, **temperature**, **speed**, **force** and **energy**. The stopping distance of the braking car is a length.

Five terms, five meanings

One measured result contains all five ideas. Take the length of a book cover: $L = (10.12 \pm 0.02)\ \mathrm{cm}$.

Term	Meaning	In $L = (10.12 \pm 0.02)\ \mathrm{cm}$
Quantity	the measurable property	length, L
Value	the number that results from the measurement	10.12
Unit	the standard the value is compared with	cm
Uncertainty	an estimate of the range within which the true value is expected to lie	$\pm 0.02\ \mathrm{cm}$
Error	measured value − true value; usually unknown; not a mistake	unknown unless the true value is known

Uncertainty is our estimate of the doubt in the value. Error is the actual difference from the true value. They are different ideas.

Operational definitions

How does a quantity get its value? By a specified procedure.

Length

Compare the object with a standard length, for example a ruler marked in metres and millimetres.

Time

Compare the interval with a standard of time, for example read a clock at the start and the end.

Mass

Compare the object with a standard mass, for example by balancing it against known masses.

Speed (derived)

There is no new procedure: measure a length and a time, then divide, speed = length ÷ time.

Rule: operational definition

An operational definition says what to do, step by step, to assign a value to a quantity.

Fundamental and derived quantities

Quantities are not all alike. Some are taken as independent, and others are defined from them.

Fundamental

Taken as independent: it is not defined in terms of other physical quantities. Its meaning is established through a specified measurement procedure (an operational definition), rather than by defining it in terms of other quantities.

Examples: length, time, mass, temperature.

Derived

Defined mathematically in terms of fundamental quantities, even though it may sometimes be measured directly.

Examples: speed, force, energy.

Physics defines a fundamental quantity by the procedure that measures it, and a derived quantity from the fundamental ones. Either way, the quantity is expressed by a numerical value and a unit.

Grouping the quantities

Here are the same quantities in one table, grouped by type.

Quantity	Type	How it is specified	SI unit
Length	fundamental	compare with the SI metre	metre, m
Time	fundamental	compare with the SI second	second, s
Mass	fundamental	compare with the SI kilogram	kilogram, kg
Temperature	fundamental	compare with a temperature scale	kelvin, K
Speed	derived	length divided by time	m/s
Force	derived	from mass and acceleration, $F = ma$ (Chapter 4)	newton, N
Energy	derived	from other quantities, for example work = force $\times$ distance (Chapter 6)	joule, J

Each fundamental quantity is fixed by comparison with a standard, and the SI units are now set by exact values of constants (Section 1.3). Every entry is expressed by a numerical value and a unit. We ask next why the unit matters.

Before any mathematics: which of these quantities needs a direction?

Scalar: size and unit

Time, mass, temperature, distance travelled and speed. Example: the car travels at $25\ \mathrm{m/s}$.

Vector: size, unit and direction

Displacement, velocity and force. Example: the car travels at $25\ \mathrm{m/s}$ to the north.

Same walk, two answers

Walk $3\ \mathrm{km}$ east, then $3\ \mathrm{km}$ west. The distance travelled is $6\ \mathrm{km}$, but you are back where you started, so the displacement is zero.

Next week we add and resolve vectors. Today we only note which quantities need a direction.

Units

How do we give a number its meaning, and how do we convert it?

KEY IDEA

A number alone means nothing: every physical quantity needs its unit.

Either kind of quantity is stated as a number, and a bare number tells you nothing.

"3.05"

Writing just "3.05" means nothing: 3.05 what?

"3.05 m"

A basketball hoop is $3.05\ \mathrm{m}$ above the ground.

Reference standard

We compare with a **reference standard**: the hoop is 3.05 times as long as a metre stick, which we define to be $1\ \mathrm{m}$ long. The standard defines the **unit**.

Rule: number and unit

A physical quantity is expressed by a numerical value together with a unit. The number alone is meaningless, so always give the unit.

SI base units

A unit comes from a reference standard, and that standard must be the same for everyone.

Quantity	Unit	Standard
Time	second, s	the time required for $9\,192\,631\,770$ cycles of the microwave radiation of a cesium-133 atom
Length	metre, m	the distance that light travels in vacuum in $1/299\,792\,458$ second
Mass	kilogram, kg	since 2018, based on Planck's constant, $h = 6.62607015 \times 10^{-34}\ \mathrm{kg \cdot m^2/s}$

SI (Système International) is used by scientists and engineers around the world. The gram is 0.001 kg. Kelvin (temperature) and mole (amount of substance) join us in the thermal physics chapters.

Derived units

Speed, force and energy need units too. All of them are built from the base units.

Quantity	Unit	Symbol	In base units
Speed	metre per second	m/s	length divided by time
Force	newton	N	$\mathrm{kg} \cdot \mathrm{m}/\mathrm{s}^2$
Energy	joule	J	$\mathrm{kg} \cdot \mathrm{m}^2/\mathrm{s}^2$
Power	watt	W	$\mathrm{J}/\mathrm{s} = \mathrm{kg} \cdot \mathrm{m}^2/\mathrm{s}^3$

Other derived units can be formed from the fundamental units: a derived unit is a product or ratio of base units.

Unit prefixes

Some units come out very large or very small, so we use prefixes. **Predict first:** *how many millimetres make one metre?*

Prefix	Symbol	Factor	Young's example
nano	n	10^{-9}	$1\ \mathrm{nm}$: a few times the size of the largest atom
micro	μ	10^{-6}	$1\ \mu\mathrm{m}$: size of some bacteria and other cells
milli	m	10^{-3}	$1\ \mathrm{mm}$: diameter of the point of a ballpoint pen
centi	c	10^{-2}	$1\ \mathrm{cm}$: diameter of your little finger
kilo	k	10^{3}	$1\ \mathrm{km}$: distance in a ten minute walk

A prefix always means the same factor, so $1\ \mathrm{km} = 10^3\ \mathrm{m}$, $1\ \mathrm{kg} = 10^3\ \mathrm{g}$ and $1\ \mathrm{m} = 10^3\ \mathrm{mm}$.

With units and standards agreed, we can use quantities in calculations. First, does an equation make sense?

Rule

Each algebraic symbol denotes both a number and a unit. Two terms may be added or equated only if they have the **same units**.

Consistent

$d = vt$, since $10\ \mathrm{m} = \left(2\ \dfrac{\mathrm{m}}{\mathrm{s}}\right)(5\ \mathrm{s})$.

Not consistent

$d = v + t$, since $\mathrm{m} \neq \mathrm{m/s} + \mathrm{s}$.

Units as symbols

Units are treated like algebraic symbols. The s in the denominator cancels, so the product has units of metres, as it must.

Habit

Carry the units through every calculation. A wrong unit means a mistake.

What if the units in a problem do not match? We convert them.

Idea

$1\ \text{min} = 60\ \text{s}$ does not say $1 = 60$. The ratio $\dfrac{1\ \text{min}}{60\ \text{s}}$ equals **1**, and so does its reciprocal.

Right factor

$$3\ \text{min} = (3\ \text{min})\left(\frac{60\ \text{s}}{1\ \text{min}}\right) = 180\ \text{s}.$$

Wrong factor

$(3\ \text{min})\left(\dfrac{1\ \text{min}}{60\ \text{s}}\right) = \frac{1}{20}\ \text{min}^2/\text{s}$, which is not a time.

Evaluate

$180\ \text{s}$ is reasonable, because the second is a smaller unit than the minute, so there are more seconds.

Habit

Work in metres, kilograms and seconds, and convert the final answer last.

Let us apply the method to a real speed.

The world land speed record is $1228.0\ \mathrm{km/h}$ (Thrust SSC). Express it in metres per second.

Predict first: will the number in m/s be larger or smaller than 1228.0?

IDENTIFY Convert km/h to m/s. We need $1\ \mathrm{km} = 1000\ \mathrm{m}$ and $1\ \mathrm{h} = 3600\ \mathrm{s}$.

SET UP
$$v = 1228.0\ \frac{\mathrm{km}}{\mathrm{h}} \times \frac{1000\ \mathrm{m}}{1\ \mathrm{km}} \times \frac{1\ \mathrm{h}}{3600\ \mathrm{s}}$$

EXECUTE
$$v = \frac{1228.0 \times 1000\ \mathrm{m}}{3600\ \mathrm{s}} = \boxed{341.11\ \mathrm{m/s}}$$

EVALUATE Smaller, as predicted. Divide km/h by 3.6 to get m/s: $90\ \mathrm{km/h} = 25\ \mathrm{m/s}$.

Check by proportion

$$1\ \frac{\mathrm{km}}{\mathrm{h}} = \frac{1000\ \mathrm{m}}{3600\ \mathrm{s}}$$

$$1228.0\ \frac{\mathrm{km}}{\mathrm{h}} = X$$

$$X = \frac{1228.0 \times 1000\ \mathrm{m}}{3600\ \mathrm{s}} = 341.11\ \mathrm{m/s}$$

Example 2: converting volume units

A volume needs more care, because the length factor is cubed.

A cut diamond has volume $30.2\ \text{cm}^3$. Express it in mm^3 and in m^3.
Predict first: will the mm^3 number be 10 or 1000 times larger?

IDENTIFY Convert cm^3 to mm^3 and to m^3.

SET UP $1\ \text{cm} = 10\ \text{mm}$ and $1\ \text{m} = 100\ \text{cm}$. Cube each factor.

EXECUTE
$$30.2\,\text{cm}^3\,(10\,\text{mm/cm})^3 = 30.2 \times 10^3\,\text{mm}^3$$
$$\boxed{30\,200\ \text{mm}^3}$$
$$30.2\,\text{cm}^3\,(1\,\text{m}/100\,\text{cm})^3 = 30.2 \times 10^{-6}\,\text{m}^3$$
$$\boxed{3.02 \times 10^{-5}\ \text{m}^3}$$

EVALUATE 1000 times, as predicted. The m^3 is a huge unit, so its number is tiny.

Check by proportion

$$\frac{1\ \text{cm}^3}{30.2\ \text{cm}^3} = \frac{10^3\ \text{mm}^3}{X}$$

$X = 30.2 \times 10^3\ \text{mm}^3 = 30\,200\ \text{mm}^3$
In m^3: $30.2 \times 10^{-6} = 3.02 \times 10^{-5}\ \text{m}^3$

Uncertainty

How far can we trust a measurement, and how do we say so?

KEY IDEA

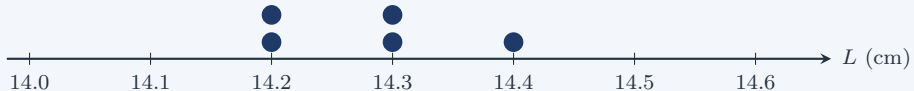
Measurement uncertainty is an estimate of the range within which the true value is expected to lie.

No measurement is exact

Units tell us what we measure. The next question is how far we can trust the number.

Five students measure the length of the same pen with the same ruler.
Predict first: will they all write down the same number?

Their readings, in cm (illustrative)



What we see

The readings are 14.2, 14.2, 14.3, 14.3 and 14.4 cm. They agree closely but not exactly.

What it means

We can never be sure a reading equals the true length. Each has an uncertainty to state.

Since no reading is exact, a result needs two parts: the value and the range around it.

Definition: measurement uncertainty

Every measurement has limited precision. Measurement uncertainty is an estimate of the range within which the true value is expected to lie. It is a likely range, not a guaranteed bound, and it depends on the measurement technique.

Written as a value plus or minus an uncertainty

$$L = (10.12 \pm 0.02)\ \text{cm}$$



How to read it

The true value is likely to lie between 10.10 and 10.14 cm (a range of 0.04 cm).

Not a mistake

An uncertainty is not a sign of carelessness. It is part of an honest result.

Sources of uncertainty

Where does the range come from? There are five places to look.

1. The instrument

Resolution: the smallest division it shows.
Calibration: whether its scale is correct.

A ruler marked in millimetres cannot directly resolve $0.1\ \mathrm{mm}$, though interpolation can estimate finer.

2. The observer

Reading the scale, reaction time, and parallax: viewing the scale from an angle.

Starting a stopwatch a moment late.

3. The method

How the measurement is set up and carried out.

Timing one swing of a pendulum instead of twenty.

4. The environment

Temperature, vibration and air draughts.

A draught moves the pan of a sensitive balance.

5. The object itself

It may not be perfectly defined.

A frayed string or a rough stone has no sharp ends to measure between.

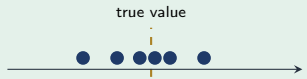
Random and systematic effects

These sources act in two different ways, and repeating a measurement treats them differently.

Random

The readings scatter about their mean, sometimes high and sometimes low. Repeating the measurement shows it.

Reaction time when stopping a stopwatch.



Systematic

Every reading is shifted the same way. Repeating the measurement does not reveal it.

Illustration, zero error: a caliper that reads $0.02\ \mathrm{mm}$ *when closed makes every reading* $0.02\ \mathrm{mm}$ *too large.*



Repeating reduces the effect of random scatter but cannot remove a systematic shift. Check the zero first: a caliper reading of $2.91\ \mathrm{mm}$ with the illustrative $0.02\ \mathrm{mm}$ zero error above is corrected to $2.89\ \mathrm{mm}$. A fuller treatment is in PHYS 105.

We cannot remove uncertainty, but we can manage it in three steps.

1. Reduce
Choose a better instrument, follow a careful procedure, control the surroundings and check the zero.

→

2. Repeat
Take several readings. Their spread shows how large the random part is.

→

3. Report
State the value, its uncertainty and the unit together, for example $(10.12 \pm 0.02)\ \mathrm{cm}$.

Next, two real instruments show how their resolution sets the size of the uncertainty.

Resolution of an instrument

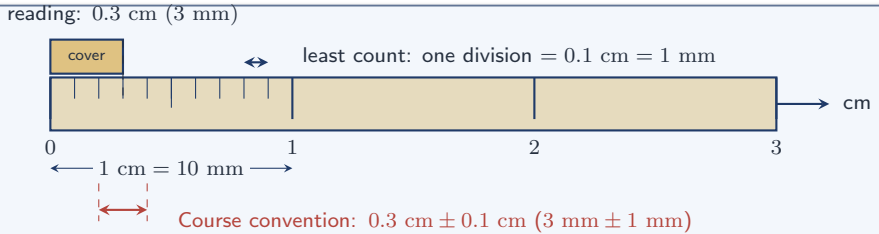
For this course we estimate the uncertainty of one reading as one least count.

Ruler

A ruler's **least count**, its smallest division, is $1\ \text{mm}$: the cover is $3\ \text{mm}$, not $3.00\ \text{mm}$.

Micrometer caliper

A micrometer caliper has a least count of $0.01\ \text{mm}$: the cover is $2.91\ \text{mm} \pm 0.01\ \text{mm}$.



Predict first: which has the smaller uncertainty, the ruler or the caliper? **Answer:** the caliper.

Resolution of common instruments

Every instrument has a least count, which sets our estimate of the uncertainty of one reading.

Instrument	Measures	Typical least count	Typical reading-uncertainty estimate under the course convention
Metre rule or ruler	length	$1\ \text{mm}$	about $\pm 1\ \text{mm}$
Vernier caliper	length	$0.1\ \text{mm}$ or $0.02\ \text{mm}$	about $\pm 0.1\ \text{mm}$ or $\pm 0.02\ \text{mm}$
Micrometer screw gauge	length	$0.01\ \text{mm}$	about $\pm 0.01\ \text{mm}$
Digital stopwatch	time	$0.01\ \text{s}$	about $\pm 0.01\ \text{s}$
Analogue clock, second hand	time	$1\ \text{s}$	about $\pm 1\ \text{s}$
Measuring cylinder	volume	$1\ \text{mL}$ or $0.1\ \text{mL}$	about $\pm 1\ \text{mL}$ or $\pm 0.1\ \text{mL}$
Liquid-in-glass thermometer	temperature	$1\ ^\circ\text{C}$ or $0.1\ ^\circ\text{C}$	about $\pm 1\ ^\circ\text{C}$ or $\pm 0.1\ ^\circ\text{C}$

Rule: uncertainty of one reading

Under the course convention the uncertainty of one reading is estimated as the least count; for a digital instrument such as the stopwatch, the last digit shown. These are typical estimates, not universal values: check your own instrument.

The uncertainty is an estimate. How does it differ from the error?

Error

The difference between a measured value and the true value: error = measured value − true value.

Illustration: true length $10.00\ \text{cm}$, measured $10.12\ \text{cm}$, so the error is $0.12\ \text{cm}$.

We usually do not know the true value, so we cannot know the error exactly. An error is not a mistake.

Uncertainty

An estimate of the range within which the true value is expected to lie.

Example: $10.12 \pm 0.02\ \text{cm}$ means the true value is likely to lie between 10.10 and $10.14\ \text{cm}$.

We never need the true value to state it, so the uncertainty is what we report.

Error is the unknown difference between the measured value and the true value. Uncertainty is our estimate of how large that difference might be. Both carry the unit of the quantity.

Every measurement has an uncertainty. How do we write it down?

Plus or minus

56.47 ± 0.02 mm: the true value is likely between 56.45 and 56.49 mm.

Fractional and percent uncertainty

$\frac{0.02 \text{ mm}}{56.47 \text{ mm}} \approx 0.0004$. Percent uncertainty $\approx 0.04\%$.

Percentage given

$47\ \Omega \pm 10\%$: about $\pm 5\ \Omega$, so between 42 and $52\ \Omega$.

Shorthand

$1.6454(21)$ means 1.6454 ± 0.0021. The digits in brackets are the uncertainty in the final digits.

Even small percent uncertainties can be very significant, for example a few metres too far on a journey of hundreds of thousands of metres.

Significant figures

Which digits of a number can we trust?

KEY IDEA

When no uncertainty is stated, the significant figures indicate, by convention, the resolution and precision of the measurement.

Predict first: *how many significant figures has* $0.250\ \mathrm{km}$*? The table shows how to count.*

Value	Sig. figs	What it tells you
$2.91\ \mathrm{mm}$	3	by convention the first two digits are taken as reliable and the third as uncertain (about $0.01\ \mathrm{mm}$)
$137\ \mathrm{km}$	3	the same number of figures, but by convention the uncertainty is about $1\ \mathrm{km}$
$0.25\ \mathrm{km}$	2	the zero to the left of the decimal point does not count
$0.250\ \mathrm{km}$	3	the zero after the decimal point counts
$1.07\ \mathrm{cm}$	3	a zero between non-zero digits counts

When an explicit uncertainty is not given, the significant figures indicate, by convention, the resolution and precision of the measurement. They are not a rigorous uncertainty: the instrument, the procedure and repeated readings set that.

Zeros at the end of a large number are ambiguous, so we change the notation.

The problem

The distance from the Earth to the Moon is about $384\,000\,000\ \mathrm{m}$. The zeros do not show the number of significant figures.

Scientific notation

Move the decimal point eight places to the left and multiply by 10^8: $3.84 \times 10^8\ \mathrm{m}$, which has three significant figures.

Zeros can count

$4.00 \times 10^8\ \mathrm{m}$ also has three significant figures: 4, 0 and 0.

Usual practice: a number between 1 and 10 multiplied by a power of 10.

Now we can write a value clearly. What happens to the digits when we calculate with it?

Rule: multiplication and division

The result has no more significant figures than the factor with the fewest.

$$\frac{0.745 \times 2.2}{3.885} = 0.42 \qquad 3.1416 \times 2.34 \times 0.58 = 4.3$$

Rule: addition and subtraction

The decimal point matters, not the count of figures.

$$123.62 + 8.9 = 132.5 \qquad 27.153 + 138.2 - 11.74 = 153.6$$

123.62 has uncertainty about 0.01, but 8.9 has about 0.1. The sum has uncertainty about 0.1, so write 132.5, not 132.52. These are reporting conventions: when uncertainties are given, they are combined separately (PHYS 105).

Rounding habits

Three habits keep the final answer reliable.

1. Round, do not truncate

$525\ \mathrm{m}/311\ \mathrm{m} = 1.688\ldots$ to three figures is 1.69, not 1.68.

2. Keep extra digits

Keep extra digits in a multi-step calculation, then round once, at the end.

3. Exact numbers

Exact numbers have no uncertainty: the 2 in $v^2 = v_0^2 + 2a(x - x_0)$ is exactly 2.

In this course

Give answers with three significant figures, or one more at most in doubtful cases. A ten-digit calculator answer overstates the precision.

Let us use the rules in a real calculation.

Find the rest energy E of an electron with $m = 9.11 \times 10^{-31}$ kg.
Predict first: how many significant figures may E have?

IDENTIFY Target: E. Known: m (3 figures) and $c = 2.99792458 \times 10^8$ m/s.

SET UP $E = mc^2$, with $1\ \text{J} = 1\ \text{kg} \cdot \text{m}^2/\text{s}^2$.

EXECUTE
$$E = (9.11 \times 10^{-31}\ \text{kg})(2.99792458 \times 10^8\ \text{m/s})^2$$
$$= 8.187659678 \times 10^{-14}\ \text{kg} \cdot \text{m}^2/\text{s}^2$$
Round to three figures: $\boxed{E = 8.19 \times 10^{-14}\ \text{J}}$

EVALUATE Three figures, as predicted. A single atom gains or loses about 10^{-19} J in a typical chemical reaction, so E is about 10^6 times larger.

Check and summary

What have we learned, and can you apply it?

KEY IDEA

Convert with unit ratios, state every result with its uncertainty, and round to the right number of digits.

Try these three questions. Decide before the answer appears.

1. A rock has mass $1.80\ \mathrm{kg}$ and volume $6.0 \times 10^{-4}\ \mathrm{m}^3$. What is its density, with the right significant figures?
   Answer: $\dfrac{1.80\ \mathrm{kg}}{6.0 \times 10^{-4}\ \mathrm{m}^3} = \boxed{3.0 \times 10^{3}\ \mathrm{kg/m}^3}$; the volume has two figures.

2. Student A states the uncertainty of a length as $0.1\ \mathrm{cm}$. Student B states the error as $0.1\ \mathrm{cm}$. Who can state the number without knowing the true length?
   Answer: Student A. The uncertainty is an estimate; the error needs the true value.

3. Two students write $15.0\ \mathrm{cm}$ and $15\ \mathrm{cm}$. Do they claim the same precision?
   Answer: No. By convention $15.0\ \mathrm{cm}$ has three significant figures and $15\ \mathrm{cm}$ has two.

Here are the rules in one table.

Idea	Rule
Physical quantity	a measurable property, expressed by a numerical value and a unit
Fundamental, derived	fundamental: independent, set by a measurement procedure; derived: defined from fundamental quantities
SI units	second, metre, kilogram as standards; derived units are products or ratios
Conversion	multiply by unit multipliers equal to 1 and let the units cancel
Consistency	add or equate only quantities with the same units
Uncertainty	state the number $\pm$ uncertainty, a likely range; for one reading the course convention is the least count; if none is given, significant figures indicate resolution by convention
Calculations	multiply or divide: fewest figures; add or subtract: decimal point

Practice questions and fuller derivations are in the tutorial and supplement, posted on Sakai.

Where this fits in the course

Physical quantities run through every block of the course.

Quantities and vectors	Motion	Forces	Energy	Momentum	Thermal physics
Chapter 1 Weeks 0 and 1	Chapters 2 and 3 Weeks 2 and 3	Chapters 4 and 5 Week 4	Chapters 6 and 7 Weeks 5 and 6	Chapter 8 Week 7	Chapters 17 to 20 Weeks 8 to 11

Every block uses the same tools: quantities with values, units and uncertainties.

Week 12 is revision and tutorials. The interim assessment covers Weeks 1 to 5; the final examination covers all content.

That is the end of this lecture. If any part is unclear, this is how to reach me.

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